moderna

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phone 617-714-6500 • (b) (6)

VIA EMAIL

November 18, 2021

(b) (6) CBER
United States Food and Drug Administration
(b) (6)
Silver Spring, Maryland 20993-0002
(b) (6) @FDA.hhs.gov

Re: FDA Site Inspection, 25-29th October 2021
Moderna Tx, Inc.
One Moderna Way
Norwood, Massachusetts 02062
FEI #: 3014937058

Dear (b) (6),

Attached please find Moderna Tx's response to the discussion with management that was held at the close of the on-site pre-licensure inspection of Moderna's Norwood manufacturing facility on October 29, 2021. This is response is being submitted in accordance with commitments made by Moderna senior leadership and consistent with Moderna's policy for addressing all health authority feedback.

The response contains specific actions that Moderna will take to address the discussion points provided by inspection team at the close-out. Moderna recognizes that the close out meeting did not reflect the full extent of the feedback provided throughout the inspection and as such, additional actions may be taken as part of our commitment to continuous improvement. Additionally, a follow up response may be submitted after review of the Establishment Inspection Report, if the need to take additional action is identified.

Moderna appreciates the collaboration with the Agency and the discussions leading up to and through the inspection and wishes to thank the Agency's inspection team for the open and constructive discussions held throughout the inspection.

For any questions or clarifications related to this response, please contact either myself at Emma.Harrington@Modernatx.com or Scott Nickerson at the contact listed below.

Kind regards,

Emma Harrington
VP, Site Head Quality Assurance, Moderna Tx. Inc., Norwood MA

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Copies (via email):

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Response to Discussion Points, 18 Nov. 2021
Moderna Tx, Inc-FDA Inspection, 25-29 Oct. 2021

Discussion Point #1

Several Moderna deviation reports involved equipment ethernet connection and network connectivity issues which will require a solution to prevent reoccurrence and assure security of the network.

Moderna Response #1:

Moderna has observed network intermittent connectivity issues between production equipment and the process control system (PCS) during manufacturing, resulting in limited data feed lapses from the local process instrumentation to the site's data historian (b) (4).

(b) (4) is a site data historian system that is used to capture and aggregate all process data generated by local automated process control systems. When an outage occurs, processing data is stored in the equipment's local control system installed on each piece of manufacturing equipment, with exception of (b) (4) have an on-board controller that monitors the (b) (4) but they do not have the capability to collect data locally. The (b) (4) readings are visually displayed on the control screen and are actively monitored by operators during the production run, as per SOP-0343, Installation and Use of (b) (4) and SOP-0324, Operation and Use of the (b) (4) (b) (4) to ensure the process is operating in a state of control. (b) (4) on-board controller which controls the process, is unaffected by the intermittent lapses. Moderna notes that the intermittent lapses affect only manufacturing equipment that feeds into the data historian, therefore facility controls and lab instruments are not similarly impacted.

Moderna initiated deviation investigations when the data feed was disrupted during batch processing. A review of the trends reveals that there were four primary root causes contributing to the lapses:

1. Resource limitations on the (b) (4) server (overloading);
2. Cable was inadvertently disconnected or a loose connection where the cable was not plugged in all the way;
3. Undetected physical damage to the cable or plug;
4. Occurrence of IP conflicts when spare ethernet connections are plugged in

Moderna has initiated several corrective actions and commits to additional actions to mitigate each of the root causes identified, as follows:

1. Upgrading the communication paths from the overloaded (b) (4) servers to industrial ethernet, Ethernet/IP under QE-007967, initiated on 03 June 2021
2. Changing the current (b) (4) ethernet connectors to industrial grade connectors to provide more rugged secure, and robust connections
3. Labeling of the (b) (4) ports to clearly identify the ports as "PCS Network Connection" and "Do Not Use"
4. Removal of cabling for any unused wireless devices which may contribute to IP conflicts when connected
5. Awareness training for operators and technicians to ensure connections are verified prior to initiation of the production process.

Commitments: Moderna commits to the following corrective actions:

- For all new expansions and new kit introductions, the digital communications network will be on the new industrial ethernet, with the upgraded industrial connectors, starting with Kit E (digital ethernet) which was rolled out in July 2021, Kit 8 (connectors and digital ethernet) by 31 December 2021, and Kit 9 (connectors and digital ethernet) which istargeted for roll-out by end of Q1 2022.

- For the other areas, barring any unforeseen supply chain concerns, Moderna will upgrade the communication and connectors on a rolling basis, aligned with the schedule for planned maintenance and shutdowns through 2022.

- The labeling and removal of spare wireless devices will be completed by December 15, 2021.

- The requirement to check the connections prior to process start and the subsequent training of operators and technicians will be completed on or before December 15, 2021.

Discussion Point #2

There are deviation reports where the root cause of the event resulted from the failure to follow procedures. Moderna should take actions to ensure procedural adherence and strengthen training of new hires.

Moderna Response #2:

Moderna has seen significant growth in volume outputs, resources, and personnel over the past year and a half in the manufacturing and QC areas resulting from the pandemic demands. This growth correlates to the increase in the number of manpower related root cause deviations where procedures were not followed, predominately across the Quality Control and Manufacturing areas. The root causes for the manpower events are mostly resulting from:

1. The need for additional clarity of some procedures
2. Introduction of new/revised procedures in a relatively short time-period
3. Onboarding of new hires across multiple shifts

To address the root causes, Moderna has allocated dedicated resources to serve as trainers in the QC Laboratories and in Manufacturing. The trainers focus on onboarding and other new hire training needs, as well as providing for expanded training opportunities beyond "read and understood" for their associates.

Within manufacturing, in addition to the role specific on the job training(OJT) that every shop floor associate is required to receive, new hires are now partnered with senior operators who coach and mentor the new hires. Additionally, Moderna has recently rolled out its "QA on the Floor" program (SOP 1267, Quality Assurance Routine Inspection of Commercial Manufacturing at Norwood) on October 7, 2021, which provides greater oversight of shop floor activities as well as serves as an additional point of contact for manufacturing and quality control associates' inquiries.

As an additional action, Moderna commits to conduct a targeted review of closed deviations from August 2021 to October 2021 in the areas of Manufacturing and QC to determine if there are additional trends or opportunities to address manpower related root cause deviations. Based on the outcome, additional actions will be taken.

In the interim, Moderna will continue to trend "manpower" related root causes and identify opportunities to improve procedures and/or training content, as well as conduct effectiveness of corrective actions taken as part of the quality management system.

Commitment:

Continue monitoring of "Manpower" quality events to identify improvement opportunities and perform review of closed deviations and initiate a Proactive Initiative within the Quality System to track actions required to reduce the number of manpower-related events.

Due Date: **20 January 2022**

Discussion Point #3:

3(a): Stability testing has not been completed as per the approved stability protocolin that there are missed stability time points. It is critical that Moderna adheres to its approved stability protocol requirements.

Moderna Response #3(a):

Moderna is manufacturing mRNA-1273 in response to the COVID-19 (SARS-CoV-2) pandemic that emerged in early 2020. Moderna received Emergency Use Authorization for its COVID-19 Vaccine by the U.S. Food and Drug Administration on December 18, 2020. Prior to the authorization, Moderna's Norwood facility served as a (b) (4) manufacturing and testing facility. The COVID vaccine was the first product commercialized by the company.

From December 2020 through the end of August 2021, Moderna has released (b) (4) million doses (corresponding to approximately (b) (4) Drug Substance and approximately (b) (4) (b) (4) Drug Product). This has led to an increase in operational difficulties within the laboratory including the timely testing of stability timepoints and is a direct consequence of the dramatic increase in the quantity of testing in the QC group.

The laboratory and general operational factors identified as contributing root causes have been assessed and Moderna has a strategic plan to remediate these issues. These actions include:

- Addition of equipment and dedicated resources to support the stability program.
- Addition of senior laboratory staff and laboratory schedulers to enable schedule adherence.
- Expansion of the laboratory footprint and equipment to increase overall capacity.

Moderna will phase these improvements through Q1 2022, in alignment with its hiring and laboratory expansion plans. Going forward, Moderna will also ensure that stability timepoints included in the agreed stability protocol for mRNA-1273 will be adhered to for the upcoming timepoints, in accordance with internal SOP's and policies. Lastly, Moderna will complete a retrospective analysis of the stability program and implement additional actions, where necessary, on or before 01 February 2022.

Commitment: Ensure timely testing and reporting of stability data according to the approved protocols for pulls after December 15, 2021.

Due Date: **December 15, 2021**

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Discussion Point #3b:

(b) (4) shelf life of (b) (4) should be revised based on available stability data until Moderna has a full data set to support (b) (4)

Moderna Response #3(b):

Moderna commits to revise the expiry in accordance with the Agency's feedback following BLA review.

Commitment: Revise shelf-life based on feedback from BLA review

Due Date: **TBD** based on Agency feedback

Discussion Point #3c:

There should be the same protocol for (b) (4) different protocols.

Moderna Response #3(c):

Stability protocols were created for (b) (4) (b) (4) (b) (4) Moderna commits to standardize its Stability Protocols across the drug substance and drug product for commercial stability studies.

Commitment: Standardize Stability Protocols

Due Dates: **28 February 2022**

Discussion Point #4

(b) (4) have good data to support performance, however Moderna should increase its data set to demonstrate that there is no (b) (4) from run to run.

Moderna Response #4:

Moderna agrees that increasing the data set will provide additional confirmation that there is no carryover from run to run. Since the inspection, Moderna has obtained additional test results for (b) (4) performance (refer to Table 5-6 in GMP-RPT-0050) that were not available at the time of the inspection. The (b) (4) (b) (4) testing continues to demonstrate acceptable (b) (4) performance.

Report GMP-RPT-0050, included in this response as Attachment 1, contains the (b) (4) data available to date. Moderna commits to (b) (4) as predicted by the (b) (4) analysis performed in GMP-RPT-0050. Although the predicted (b) (4) (b) (4) Moderna will conservatively restrict the (b) (4) to (b) (4) basedon the data. It is expected that the (b) (4) (b) (4) will reach the targeted endpoint in (b) (4) (b) (4) respectively. After reaching the final endpoints, the remaining data assessed, and the final report prepared demonstrating the allowable (b) (4)

Commitment:

Moderna will finalize a final (b) (4) report after completion of the testing and analysis of the (b) (4)

Due Date:

The final report is targeted for completion by **31 March 2022.**

Discussion Point #5

Comments in laboratory worksheets must be supported by documentation, objective evidence and/or supporting data or reference to a deviation.

Moderna Response #5:

Moderna has conducted an extensive review of its lab operations and revised its procedures for handling of Laboratory Events, SOP 0409, effective July 20, 2021. The procedure outlines the requirements for addressing unexpected data results and defines the documentation requirements to support comments and conclusions. Prior to this revision, there were instances identified where the documentation was not available or complete to support comments on the lab sheets. This prompted the revision of the procedure, additional training on good documentation practices, and increased oversight to help coach and mentor laboratory staff on GMP laboratory practices during the months of August and September 2021.

Moderna also identified the need to update the data review procedure, SOP 0082, Data Review and Reporting in the GMP Quality Control Laboratory, to provide additional guidance to data reviewers on what and how to verify all supporting documentation and evidence when assessing a data package. This procedure is in the process of being revised and is expected to be finalized by December 15, 2021. The procedure will also require that there be the appropriate cross-references between quality event records and comments on laboratory worksheets.

Moderna will also develop a specialized data reviewer training and certification program to further strengthen the data review skill sets and drive additional consistency in data review. In the interim, additional quality assurance oversight has been implemented to support the laboratory operations and ensure the completeness of the data packages.

Commitments:

1. Revision to SOP 0082 to provide additional guidance for data reviewers
Due Date: **15 December 2021**

2. Development of Specialized Data Reviewer Training and Certification
Due Date: **31 January 2022**

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Response to Discussion Points, 18 Nov. 2021
Moderna Tx, Inc-FDA Inspection, 25-29 Oct. 2021

Discussion Point #6

The (b) (4) leak investigation requires additional actions such as performing a qualification of the (b) (4) components, implementing an AQL sampling at incoming QC inspection, and making the operator training more robust and ensuring procedures have more detailed instructions for (b) (4) handling/installation.

Moderna Response #6

Although the (b) (4) leak rate has steadily fallen (from 1.9% to 0.4%) since the implementation of several CAPAs during the February to May 2021 timeframe, Moderna is committed to pursue a number of engineering, operational, and quality system improvements intended to further reduce the incidence rate. As outlined in the investigation, QE 004886 there are seven (7) active CAPAs that will address specific failure modes and root causes identified. These include:

1. Improvements to procedures and training, including step-by-step training videos for (b) (4) opening, inspection, and installation.
2. Formal written procedures for responding to (b) (4) leak events.
3. Design change to remove unused (b) (4) eliminating leaks due to (b) (4) (b) (4)
4. Implementation of (b) (4) to minimize abrasion leaks in collapsible (b) (4)
5. Improvements to (b) (4) configuration for easier cleaning and easier (b) (4) installation.
6. Improvements to the design and arrangement of storage areas, shelving, racks, and carts.
7. Improvements to written procedures for the storage and transport of (b) (4)

Additionally, Moderna recognizes that there were elements of the (b) (4) quality program that require further enhancement, including formal qualification of (b) (4) inspection procedures for (b) (4) and a comprehensive program for training and education on the (b) (4) handling, storage, inspection, installation and use. To this end, Proactive Initiative (PI) QE-011673 was opened on 28 October 2021 to address the following:

1. Material Qualification (Material Comparability, Design Qualification, Manufacturing Quality, and Confirm Vendor Certification)
2. (b) (4) Inspection (QC Incoming Inspection and Pre-Use Inspection)
3. Training and Qualification of Operators on the handling and storage of (b) (4)

Moderna will continue to monitor the incidents of (b) (4) leaks and the effectiveness of the corrective actions taken as these improvements are rolled out.

Commitments:

1. **Material Qualification (Material Comparability, Design Qualification, Manufacturing Quality, and Confirm Vendor Certification)**

   The standard operating procedure (SOP) and qualification master plan (QMP) for (b) (4) (b) (4) will be approved in QMS (b) (4) no later than **31 January 2022.** The master plan will include target dates for the completion for qualification each (b) (4) part number.

2. (b) (4) **Inspection (QC Incoming Inspection and Pre-Use Inspection)**

   The standard operating procedure for incoming QC inspection of (b) (4) will be approved in QMS (b) (4) no later than **31 January 2022** The standard operating procedure(s) for pre-use inspection of (b) (4) will be approved in QMS (b) (4) no later than **31 January 2022.**

3. **Training and Qualification of Operators on the handling and storage of** (b) (4)

   The standard operating procedure for the storage, transport, handling, and use of (b) (4) (b) (4) and the related training materials will be approved in QMS (b) (4) no later than **31 January 2022**. The initiation of an on-the-job training program, which will include an end-user qualification, will begin immediately thereafter. At that point, all employees that are new to their role will be required to complete the OJT / end-user qualification prior to performing routine activities. A timetable for qualification of existing employees, as well periodic requalification will be included in the training program.

Due Date: January 31, 2022

Attachment 1

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